CHANDAN SURESH

AI Infrastructure Validation & NPI Engineer | GPU Platform Bring-Up, Production Readiness & Automation

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PROFESSIONAL SUMMARY

Production systems and validation engineer with 5+ years across large-scale AI data center infrastructure, hardware/firmware debug, and automation. NVIDIA-certified in AI Infrastructure and Operations. Currently lead validation engineer for NVIDIA VR200 (Vera Rubin) rack-scale NPI — DVT complete, PVT in progress — after owning GB200/GB300 NVL72 bring-up, L10/L11 validation, and deployment readiness for hyperscale (Meta) data center rollouts. Hands-on across liquid-cooled rack integration, NVLink/NVSwitch and PCIe fault isolation, BMC/BIOS/CPLD firmware install, recovery and validation, hardware telemetry and Linux diagnostics, and Arista/NVIDIA switch troubleshooting. Builds Python and Bash automation for qualification, health checks, log analysis, and regression coverage, and works directly with NVIDIA platform teams and hyperscale customers through qualification, escalation, root-cause analysis, and corrective action.

CORE SKILLS

GPU Platforms & NPI: NVIDIA VR200 / Vera Rubin rack-scale NPI (DVT, PVT), GB200 & GB300 NVL72, compute and NVLink switch trays, Grace CPUs, Blackwell GPUs, rack-level bring-up, L10 & L11 validation, rework racks, EVT/DVT/PVT gate criteria, production readiness & operational handoff

Systems & Hardware: Linux administration & troubleshooting, server architecture, BIOS/UEFI, BMC, IPMI/Redfish, CPLD, firmware install/recovery/validation, PCIe, NVMe, NICs, NVSwitch/NVLink link diagnostics, PSU, thermal & sensor debug, hardware telemetry, failure diagnostics, RMA & corrective action

Data Center Infrastructure: High-density rack integration, liquid-cooled GPU racks, power delivery, structured cabling, network fabric validation, deployment safety, hyperscale deployment readiness

Networking: Arista and NVIDIA switch troubleshooting, link/connectivity fault isolation, fabric cabling validation, platform instability debug

Automation & Programming: Python, Bash/shell scripting, C++, C#, C, Java, SQL (MySQL, SQL Server), Git, CI/CD, test suite & framework development, log/telemetry parsing, automated health checks & regression testing

AI-Assisted Engineering: Applying LLM tooling to failure-log triage, troubleshooting, knowledge retrieval, documentation, and script/code generation

Quality & Process: Validation strategy & planning, system integration testing, root cause analysis, ISO & IEEE testing standards, black box & white box testing, defect tracking (Jira), KPI & yield tracking, Agile

Certifications: NVIDIA AI Infrastructure and Operations (Jun 2026); Software Testing Bootcamp (2025, Udemy & Tosca); ISTQB Foundation Level (in progress)

PROFESSIONAL EXPERIENCE

Test Engineer II — AI Systems Validation & NPI | Quanta Computer *Aug 2026 – Present*

- Lead validation engineer for NVIDIA VR200 (Vera Rubin) rack-scale AI platform NPI: completed DVT and currently driving PVT, owning test execution, failure triage, and exit-criteria readiness ahead of mass production.
- Define and execute rack-scale qualification and burn-in coverage across compute trays, NVLink switch trays, power delivery, cooling, cabling, and rack management, ensuring next-generation platforms are production- and operations-ready at handoff.
- Own root-cause analysis for high-impact hardware and firmware failures involving GPUs, NVSwitch, CPLD, PSUs, sensors, and thermals; coordinate mitigation and corrective action with design, manufacturing, and NVIDIA platform teams, and verify fixes through regression validation.
- Build and maintain Python and Bash automation for validation coverage, hardware telemetry and low-level log analysis, firmware workflows, and health checks, cutting manual test and debug cycle time.
- Establish repeatable validation procedures, debug playbooks, and technical documentation; mentor test engineers and technicians on rack bring-up, firmware recovery, and cross-layer troubleshooting methodology.
- Apply LLM-based tooling to accelerate failure-log triage, knowledge retrieval, and script generation across validation workflows.

Test Engineer | Quanta Computer *Sep 2025 – Aug 2026*

- Delivered validation and deployment readiness for NVIDIA GB200 and GB300 NVL72 AI infrastructure supporting Meta hyperscale data center deployments and New Product Introduction (NPI) programs.
- Performed platform bring-up, rack-level validation, L10 and L11 testing, and rework across large-scale liquid-cooled GPU rack environments.

- Debugged complex system-level failures spanning GPUs, NVSwitch, CPLD, sensors, PSUs, and firmware at both rack and node level, isolating faults across hardware, firmware, and OS layers.
- Troubleshot Arista and NVIDIA switches to resolve hardware communication issues, link and connectivity failures, and platform instability.
- Analyzed hardware telemetry, BMC/system logs, and Linux-based diagnostics using Bash and Python to identify root causes and drive measurable reliability improvements.
- Executed firmware installation, recovery, validation, and troubleshooting across NVIDIA AI compute platforms and management controllers.
- Developed and maintained test suites and supporting automation to expand validation coverage and improve testing efficiency; collaborated cross-functionally with engineering, manufacturing, and validation teams through all NPI deployment phases.

Volunteer Software Developer | BrightMind Enrichment & Schooling *Aug 2024 – Present*

- Built performance dashboards tracking test coverage, defect trends, and issue-resolution time, turning QA data into actionable reporting for stakeholders.
- Analyzed recurring defect patterns and proposed corrective actions; performed API validation with Postman and REST-assured, handling confidential project data with discretion.

Teaching Assistant | Cleveland State University *Aug 2022 – May 2024*

- Led 150+ hours of coding lab sessions and one-on-one support for 200+ students on data structures, algorithms, debugging, and software development practices; authored instructional documentation for complex technical concepts.

Application Development Engineer | Accenture *Jan 2021 – Aug 2022*

- Executed functional, integration, and unit validation (black box and white box) against software specifications, authoring 10,000+ test scenarios, cases, and defect/summary reports across all SDLC phases.
- Optimized test scripts and led code reviews, improving test execution time by 30%; automated reporting to surface process inefficiencies in UAT cycles.
- Managed defect tracking and issue resolution to validation standards and provided troubleshooting support for internal and client applications.

SELECTED PROJECT

Smart Pharmaceutical Refrigerator (IoT) — *Python*, *Arduino Uno*, *OpenCV*, *TensorFlow*, *CNNs* | *May 2023*

- Engineered an IoT-monitored pharmaceutical refrigerator with sensor telemetry and CNN-based object detection for inventory tracking and real-time low-stock/misplacement alerts, improving detection accuracy by 92% and reducing manual intervention by 80%.

EDUCATION

Master of Science, Computer Science — Cleveland State University, Cleveland, OH | Aug 2022 – May 2024

Bachelor of Engineering, Computer Science — PES College of Engineering, India | Aug 2016 – Aug 2020